CI/CD Pipeline Diagram Template—Enterprise GenAIOps Platform

Version 2026.1 | Azure DevOps & GitHub Actions Reference Architecture | LC 3 Project Module

Template Overview

This template provides a standardized visual framework for documenting your CI/CD pipeline architecture. It aligns with Microsoft Azure DevOps 2026, GitHub Actions, Azure AI Foundry, and Azure API Management patterns. Use the diagram structure below as a starting point, then customize per-application gates, thresholds, and approval flows.

1. Pipeline Stage Reference Architecture

The following stages represent the recommended minimum viable pipeline for enterprise GenAI deployments. Each stage includes inputs, outputs, decision gates, and Azure service mappings.

1.1 Stage Definitions

Stage	Purpose	Azure Services	Gate Type	Output Artifacts
1. Source and Trigger	Code commit, PR validation, branch protection	GitHub / Azure Repos, Branch Policies, GitHub Actions triggers	Manual (PR review)	Merged code, commit SHA
2. Build and Package	Compile orchestration logic, containerize apps, package prompts/configs	Azure Container Registry, GitHub Actions / Azure Pipelines, Bicep/Terraform	Automated (build success)	Container images, Bicep modules, prompt bundles
3. Model Validation	Validate model artifacts, version compatibility, prompt-to-model fit	Azure AI Foundry Model Catalog, Azure ML Model Registry	Automated (model signature check)	Validated model reference, compatibility report
4. Automated Evaluation	Run quality gates: unit tests, integration tests,	Azure AI Evaluation SDK, Prompt Flow,	Automated (threshold-based)	Evaluation report, pass/fail per metric

	LLM-specific metrics	Azure Load Testing		
5. Artifact Packaging	Bundle model + prompts + configs + infra-as-code into versioned release	Azure Artifacts, GitHub Packages, semantic versioning	Automated (packaging validation)	Release bundle (for example, v2.3.1-askretail)
6. Staged Promotion	Deploy to Dev → Staging → Production with environment-specific configs	Azure DevOps Environments, GitHub Environments, Azure App Config	Manual + Automated (approval gates)	Deployed application, environment state
7. Post-Deploy Validation	Smoke tests, synthetic traffic, canary analysis, rollback readiness check	Azure Monitor, Application Insights, Azure Load Testing	Automated (health checks)	Deployment health report, telemetry baseline

2. Visual Diagram Structure (Text-Based Template)

Below is the ASCII/Mermaid-style diagram template you should adapt in your diagramming tool (draw.io, Lucidchart, Visio). Callouts [C1]–[C12] map to explanatory notes in Section 3.

Mermaid Diagram Source (for draw.io / Markdown rendering):

```
graph TD
    A[Developer Push/PR] -->|Trigger| B[Build & Package Stage]
    B --> C[Model Validation Gate]
    C -->|Pass| D[Automated Evaluation Stage]
    C -->|Fail| X[Reject & Notify]
    D -->|Pass| E[Artifact Packaging]
    D -->|Fail| X
    E --> F[Dev Environment Deploy]
    F --> G[Dev Smoke Tests]
    G -->|Pass| H[Staging Approval Gate]
    G -->|Fail| X
    H -->|Approved| I[Staging Deploy]
    I --> J[Integration & Load Tests]
    J -->|Pass| K[Production Approval Gate]
    J -->|Fail| X
    K -->|Approved| L[Progressive Rollout]
    L --> M[Canary / Blue-Green Traffic Split]
    M --> N[Post-Deploy Validation]
```

```
N -->|Healthy| O[Full Traffic Cutover]
N -->|Degraded| P[Auto-Rollback Trigger]
P --> Q[Previous Version Restoration]
O --> R[Production Monitoring Baseline]

subgraph "AskRetail (Tier 1)"
    D1[Latency < 800ms p50]
    D2[Groundedness ≥ 0.92]
    D3[Hallucination < 1.5%]
    D4[Security Scan: Critical=0]
end

subgraph "StockSense (Tier 2)"
    D5[MAPE < 8%]
    D6[Batch Completion < 45min]
    D7[Integration Test Pass]
end

subgraph "HRHelper (Tier 1-S)"
    D8[Content Safety: Hate/Self-Harm=0]
    D9[PII Detection: Pass]
    D10[Audit Trail Integrity Check]
    D11[Escalation Accuracy ≥ 95%]
end

D --> D1 & D2 & D3 & D4
D --> D5 & D6 & D7
D --> D8 & D9 & D10 & D11
```

3. Diagram Callout Explanations

[C1] Source and Trigger: Protect main branch with required reviewers (2+ for Tier 1, 1+ for Tier 2), status checks, and signed commits. Use GitHub Actions `workflow_dispatch` for manual triggers and `push`/`pull_request` for automated triggers.

[C2] Build and Package: Build container images with Azure Container Registry Tasks or GitHub Actions. Tag images with Git commit SHA and semantic version. Package prompts as immutable JSON/YAML bundles versioned alongside code.

[C3] Model Validation Gate: Verify model artifact exists in Azure AI Foundry Model Catalog. Check prompt-to-model compatibility matrix (for example, GPT-4o requires prompt schema v2.1+). Reject if "latest" tag is detected in production-bound configs.

[C4] Automated Evaluation—AskRetail: Run Azure AI Evaluation SDK groundedness, relevance, coherence, and fluency checks against a held-out test set (min 200 samples). Latency test via Azure Load Testing: 500 concurrent users, 5-minute duration. Security scan with Microsoft Defender for Cloud and Trivy.

[C5] Automated Evaluation—StockSense: Validate forecast accuracy against 30-day holdout period. Integration tests cover SAP ERP connector and Azure Synapse query paths. Batch performance test: simulate 10,000 SKUs end-to-end.

[C6] Automated Evaluation—HRHelper: Content safety via Azure AI Content Safety API (Prompt Shield, Protected Material Detection). PII detection via Microsoft Presidio or Azure AI Language PII extraction. Audit trail: verify all interactions log to immutable Azure Blob Storage with WORM policy.

[C7] Artifact Packaging: Use semantic versioning (MAJOR.MINOR.PATCH). Release bundle structure: `release-{app}-{version}.zip` containing: (1) container image digest, (2) prompt bundle manifest, (3) Bicep/Terraform templates, (4) model reference JSON, (5) evaluation report summary.

[C8] Environment Promotion: Dev: auto-deploy on green build. Staging: requires approval from App Owner (for example, CSCO for StockSense). Production: requires approval from VP Platform Engineering + App Owner for Tier 1; additional AI Ethics Committee sign-off for Tier 1-S (HRHelper).

[C9] Progressive Rollout: AskRetail: Blue/Green with 5-minute health window. StockSense: Canary 10% → 50% → 100% over 2 hours. HRHelper: Blue/Green with extended 30-minute observation due to sensitivity.

[C10] Post-Deploy Validation: Synthetic traffic via Azure Monitor Application Insights Availability Tests. Compare error rate, latency p99, and groundedness score against pre-deployment baseline. Auto-rollback if deviation > 10% for Tier 1, > 20% for Tier 2.

[C11] Rollback Mechanics: Switch traffic at API Management layer (instant for blue/green, weighted for canary). Revert configuration in Azure App Configuration. Re-deploy previous artifact bundle. Notify on-call via PagerDuty/Opsgenie integration.

[C12] Parallel vs. Shared Steps: Shared: Build, packaging, infra provisioning. Parallel: App-specific evaluation gates, staging deployments, and production rollouts. Show parallel swimlanes per app in your diagram.

4. Approval Matrix Template

Environment	AskRetail (Tier 1)	StockSense (Tier 2)	HRHelper (Tier 1-S)

Development	Auto-deploy on green build	Auto-deploy on green build	Auto-deploy on green build
Staging	App Owner (CDO delegate) + Platform Eng	App Owner (CSCO delegate)	App Owner (CHRO delegate) + Compliance Officer
Production	VP Platform Eng + CDO delegate + On-call SRE	VP Platform Eng + CSCO delegate	VP Platform Eng + CHRO delegate + AI Ethics Committee + Legal Counsel

5. Evaluation Gate Threshold Template

Use this table to define your per-application, per-metric thresholds. All thresholds must be numerically specified.

Metric	AskRetail Threshold	StockSense Threshold	HRHelper Threshold	Gate Action
Unit Test Coverage	≥ 85%	≥ 70%	≥ 80%	Block if below
Integration Test Pass Rate	100%	≥ 95%	100%	Block if below
Groundedness Score	≥ 0.92	N/A (retrieval N/A)	≥ 0.90	Block if below
Hallucination Rate	< 1.5%	N/A	< 2.0%	Block if above
Latency p50	≤ 800ms	≤ 8s (ad-hoc)	≤ 3.5s	Warn if exceeded; block if >120%
Latency p99	≤ 2.5s	≤ 15s (ad-hoc)	≤ 6s	Block if above
Error Rate	< 0.1%	< 1%	< 0.5%	Block if above
Security Scan (Critical)	0	0	0	Block if >0
Content Safety Violations	0 (Hate/Violence)	N/A	0 (all categories)	Block if >0
Token Cost per Request	≤ $0.04	≤ $0.12 (complex)	≤ $0.08	Warn if exceeded

6. Rollback Trigger Template

Trigger Condition	Auto/Manual	Detection Source	Rollback Action
Post-deploy health check failure	Auto	Application Insights Availability Tests	Traffic switch + artifact revert

Error rate spike > 5× baseline	Auto	Azure Monitor Alert Rule	Traffic switch + notify on-call
Latency p99 > SLA × 1.5 for > 5 min	Auto	Application Insights Performance Counters	Progressive traffic reduction → full rollback
Groundedness score drop > 0.05	Manual	Azure AI Evaluation continuous runner	Model version revert + prompt rollback
Hallucination rate > 3% in canary	Auto	Custom LLM metric exporter	Canary halt + previous version restore
Security incident (data exfiltration)	Manual	Microsoft Sentinel / DLP alerts	Immediate full shutdown + forensic preservation
Cost per request > 200% of budget	Auto	Azure Cost Management + API Management metrics	Throttling + quota reduction + notify CFO

7. Microsoft Reference Alignment (2026)

- Azure DevOps 2026: Use multi-stage YAML pipelines with environment approvals and checks. Leverage Azure Deployment Environments for ephemeral dev/test environments.
- GitHub Actions: Use reusable workflows for shared build/test steps. Implement OIDC federation with Azure for secure, secret-less authentication.
- Azure AI Foundry: Register all models in the Model Catalog. Use prompt flow for evaluation orchestration. Version prompts via Git with immutable artifact references.
- Azure API Management: Configure CI/CD via Azure Resource Manager (ARM) or Bicep templates. Use named values and policies stored in Git for version control.
- GitOps (Optional): Consider Azure Arc-enabled Kubernetes with Flux v2 for automated cluster-state reconciliation, aligning with Microsoft's recommended GitOps pattern for AKS.

References: Azure DevOps Documentation 2026, GitHub Actions Reusable Workflows, Azure AI Foundry CI/CD Guide, Azure API Management DevOps Resource Kit, Microsoft Well-Architected Framework—Operational Excellence Pillar.